

EvoState: Exploring Long-Horizon Evolving States & Inference-Time Scaling

A Scientific Exploration of Memory Compression, Interference, and Test-Time Latent Deliberation

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CENTRAL FALSIFIABLE SCIENTIFIC CLAIM:

"A fixed-size evolving state can carry useful information across sequences without storing every previous token, but increasing sequence length and conflicting updates can cause interference and information loss; additional inference-time computation can sometimes improve recovery."

1. Problem & Motivation: The Context Memory Bottleneck

Modern autoregressive Transformer language models rely on the Key-Value (KV) cache to maintain exact historical token representations across context windows. While lossless, this strategy incurs an $O(T \cdot d)$ spatial memory footprint and an $O(T)$ step latency during generation. In deployment settings exceeding 100k+ tokens, VRAM consumption explodes, and generation becomes heavily memory-bandwidth bound. Alternative architectures—including state-space models (Mamba), recurrent units, and brain-inspired evolving states—compress history into an $O(1)$ bounded state. However, compressing unbounded sequence history into finite-dimensional state spaces creates fundamental trade-offs: superposition interference, finite capacity limits, and recency bias. EvoState is an educational research laboratory built to systematically dissect these trade-offs, test memory limits, and evaluate whether test-time compute can mitigate degradation.

2. Conceptual Background & Mathematical Mechanics

In our educational associative memory framework, key-value associations are folded into an evolving fast-weight state matrix $M_t \in \mathbb{R}^{(d_v \times d_k)}$ via outer-product updates:

State Update: $M_t = (1 - \Delta_t) M_{t-1} + \Delta_t (v_t \otimes k_t^T)$

Single-Pass Query Projection: $v^* = M_T^T q$

Iterative Test-Time Deliberation: $q^{(k+1)} = (1 - \eta) q^{(k)} + \eta M_T^T \sigma(q^{(k)})$

When key vectors are mutually orthogonal, retrieval is exact. As the number of stored associations N exceeds the associative rank threshold (Hopfield limit $\alpha_c \approx 0.14 d$), crosstalk causes graceful degradation. Selective gating (Δ_t) allows the state to freeze decay over silent padding or noise tokens, whereas test-time latent deliberation acts as an associative de-noising filter.

3. Architectural Comparison Across Sequence Modeling Paradigms

Architecture	State Memory	Step Latency	Interference Behavior	Primary Citation
Transformer (KV Cache)	$O(T \cdot d)$ Unbounded	$O(T)$ per token	Lossless non-parametric lookup	Vaswani et al. (2017)
Selective SSM (Mamba)	$O(d)$ Bounded	$O(1)$ Constant	Selective Δ_t gating; decay over lag	Gu & Dao (2023)
BDH: The Dragon Hatchling	$O(1)$ Bounded State	$O(1)$ Constant	Evolving internal state & plasticity	Kosowski et al. (2025)
BDH-CQ (Latent Reason)	$O(1)$ Bounded State	$O(K)$ Variable	Test-time latent state deliberation	Engdahl et al. (2026)

4. Experiment Design & Empirical Sweeps (2,700 Seeded Trials)

To test each clause of the central claim under falsifiable conditions, we executed 2,700 controlled synthetic trials across 50 distinct random seeds ($N=50$ per condition):

- EXP-1: Delayed Recall (Lag $k \in [4, 128]$):** Evaluates memory retention across neutral padding. The educational evolving state maintained 100.0% accuracy across all lags via selective gating, whereas non-selective vector recurrence decayed to 0.0%.
- EXP-2: Sequence Clutter ($T \in [32, 512]$):** Full-history baseline maintained 100% at a latency cost growing from 7.42ms to 61.11ms. Evolving state maintained $O(1)$ latency (38–70ms) with mild noise accumulation.
- EXP-3: Destructive Overwrites ($N_{\text{overwrites}} \in [1, 8]$):** Recency dominance observed; overwrite saturation dropped accuracy from 100% ($N \leq 2$) to ~45% ($N \geq 5$).
- EXP-4: Subspace Capacity Stress (N_{pairs} vs $d=32$):** A sharp phase transition occurs at $N > 4$ pairs, matching theoretical associative matrix rank bounds.
- EXP-5: Inference-Time Recovery:** Additional test-time compute ($K=1$ to 16) raised retrieval accuracy (+27.6% under moderate noise)

when the signal was in superposition, but failed to recover state that had been completely overwritten.

5. Interactive Learner Journey & Telemetry

EvoState provides a comprehensive interactive web application featuring three exploration modes:

1. **60-Second Experiment:** A rapid 5-step interactive protocol where learners adjust sequence length, capacity, noise, and inference effort, followed by an unmanipulated verdict on the central claim.
2. **Guided Pathway Lab:** A 7-stage didactic curriculum with embedded concept check questions.
3. **Open Workbench:** An unconstrained 4-parameter sandbox (L, M, p, K) paired with live D3 energy landscape visualizers, state norm monitors, and coordinate heatmaps.

6. Relationship to BDH (The Dragon Hatchling) & BDH-CQ

BDH Primary Foundation: *The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain* (Kosowski et al., 2025; arXiv:2509.26507) is a biologically inspired architecture exploring evolving internal associative states and local synaptic plasticity dynamics.

BDH-CQ Primary Foundation: *BDH-CQ: In-Context Learning with Recurrent Latent Reasoning* (Engdahl et al., 2026; arXiv:2608.09888) explores combining in-context learning with recurrent latent reasoning and test-time deliberation.

EvoState Pedagogical Role: EvoState does NOT implement official multi-billion parameter BDH weights. Instead, EvoState provides an educational simplified model inspired by the broader idea of an evolving internal state, allowing learners to observe persistence, interference, capacity limits, and recovery under controlled experiments.

7. Scientific Limitations & Evidence Classification

CRITICAL BOUNDARIES & EVIDENCE CLASSIFICATION:

1. **Educational Model vs Published BDH:** Our browser toy is an isolated d=32 pedagogical surrogate running fast-weights.
2. **Information Irreversibility:** Test-time deliberation cannot reconstruct information completely overwritten or zeroed from the state.
3. **Evidence Taxonomy:** Every claim is tagged as [PUBLISHED RESULT], [INDEPENDENT RESULT], [OUR EXPERIMENT], or [EDUCATIONAL SIMPLIFICATION].
4. **Zero Fabricated Data:** All 2,700 benchmark data points are deterministically reproducible via PyTorch across fixed seeds.

8. Verified Primary References

[1] Kosowski, A., Uznański, P., Chorowski, J., Stamirowska, Z., & Bartoszkiewicz, M. (2025). *The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain*. arXiv:2509.26507.

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